Lab Assignment:

1. To accept students five courses marks and compute his/her result. Student is passing if he/she scores marks equal to and above 40 in each course.

If student scores aggregate greater than 75 percentage, then the grade is distinction.

If aggregate is greater than or equal to 60 and less than 75 then the grade if first division.

If aggregate is greater than or equal 50 and less than 60, then the grade is second division.

If aggregate is greater than or equal 40 and less than 50, then the grade is third division.

2. Solve the Fibonacci sequence using recursive function in Python.
3. Write a Python program to print different patterns.

Student Result + Grade

```
marks = []

print("Enter marks of 5 subjects:")
for i in range(5):
    m = float(input(f"Subject {i+1}: "))
    marks.append(m)

# Check pass condition
if all(m >= 40 for m in marks):
    total = sum(marks)
    percentage = total / 5

    print("Percentage =", percentage)

    if percentage > 75:
        print("Grade: Distinction")
    elif 60 <= percentage < 75:
        print("Grade: First Division")
    elif 50 <= percentage < 60:
        print("Grade: Second Division")
    elif 40 <= percentage < 50:
        print("Grade: Third Division")

else:
    print("Result: Fail")
```

Output

```
Enter marks of 5 subjects:
Subject 1: 86
Subject 2: 56
Subject 3: 67
Subject 4: 84
Subject 5: 93
Percentage = 77.2
Grade: Distinction
```

Fibonacci Using Recursion

```
def fibonacci(n):
    if n == 0:
        return 0
    elif n == 1:
        return 1
    else:
        return fibonacci(n-1) + fibonacci(n-2)

n = int(input("Enter number of terms: "))

print("Fibonacci Series:")
for i in range(n):
    print(fibonacci(i), end=" ")
```

Output

```
Enter number of terms: 5
Fibonacci Series:
0 1 1 2 3
```

Pattern Printing
a)Right Angled Triangle

```
n = int(input("Enter rows: "))

for i in range(1, n+1):
    print("*" * i)
```

Output

```
Enter rows: 4
*
**
***
****
```

b) Inverted Triangle

```
Enter rows: 7
*******
******
*****
****
***
**
*
```

c) pyramid

```
n = int(input("Enter rows: "))

for i in range(1, n+1):
    print(" "*(n-i) + "*"*(2*i-1))
```

Output

```
Enter rows: 6
     *
    ***
   *****
  *******
 *********
***********
```